14-7-26

Assignment - 1

1. Write the Desciption of Audino Bot?

A:- Ardino Bot is a next generation AI assistant built to simplify everyday talks through intelligent automation and natural conversation. wheather you instant answers, Content creation workflow assistance.

Language Reference:-

Digital I/O :-

digitalRead():- Reads the value from a specified digital pin

syntax - digitalRead(pin).

digitalWrite():- Writes the value from a Specified digital pin.

syntax - digitalWrite(pin, value).

PinMode():- Configures the specific pin to behave either as Input or Output.

Syntax - pinMode(pin, mode).

Math :-

abs() :- Calculates the absolute value of given number.

Syntax- abs(x)

Constrain() :- Constraints a number to be within the range.

syntax - Constraint(x, a, b).

map() :- Re-maps a number from one range to another.

syntax - map(value, fromLow, fromHigh, toLow, toHigh).

Max() :- Finds the larger of two numbers.

syntax - max(x,y).

Min() :- Finds the smaller of two number ~~using~~

syntax - min(x,y)

Pow() :- Calculates the value of a number raised to a power.

Syntax - pow(base, exponent).

sq() :- Calculates the Square of a number.

syntax - sq(x).

sqrt() :- Calculates the square root of a number.

syntax - sqrt(x)

Bits and Bytes :-

bit() :- Creates a bit mask with a single set at a specific position

syntax - bit(n).

bitClear() :- Clears a bit of a numeric variable at a specific position.

syntax - bitClear(x, n).

bitRead() :- Reads a bit of a variable on a specific position.

syntax - bitRead(x, n).

bitSet() :- Sets a bit of a numeric variable at a specific position.

syntax - bitSet(x, n).

bitWrite() :- Writes to a bit of a variable

syntax - bitWrite(x, n, b).

highByte():- Extracts the high 8 bits from a 16-bit integer or the second lowest byte of a larger data type.

Syntax - highByte(x).

lowByte():- Extracts the low 8 bits from a larger data type.

Syntax - lowByte(x)

Analog I/O :-

- analogRead():- Reads the value from a specified analog input pin.

  syntax - analogRead(pin).

analogReadResolution():- this function is an extension of the ~~Anog~~ Analog API that let you set the resolution of a value.

syntax - analogReadResolution(bits)

analogReference():- Configures the reference voltage used for analog input.

syntax - analogReference ~~(pin,)~~ (type)

analogWrite():- Writes an analog value to a pin.

syntax - analogWrite(pin, value)

analogWriteResolution():- It is an extension of Analog

syntax - analogWriteResolution(bits)

Trigonometry :-

cos() :- Calculate the cosine of an angle.

Syntax - :cos(rad)

sin() :- Calculates the sine of an angle.

Syntax - sin(rad)

tan() :- Calculates the tangent of an angle.

Syntax - tan(rad)

External Interrupts :-

attachInterrupt() :- Interrupts help make things happen automatically in microcontroller programs.

syntax - attachInterrupt(digitalPinToInterrupt(pin), ISR, mode)

detachInterrupt() :- This function turns off a given interrupt that was attached previously.

Syntax - detachInterrupt(digitalPinToInterrupt(Pin)).

digitalPinToInterrupt() :- function takes a pin as an argument, and returns the same pin.

Syntax - digitalPinToInterrupt(Pin).

Advanced I/O :-

noTone () :- stops the generation of a square wave triggered by tone().

Syntax - noTone (pin).

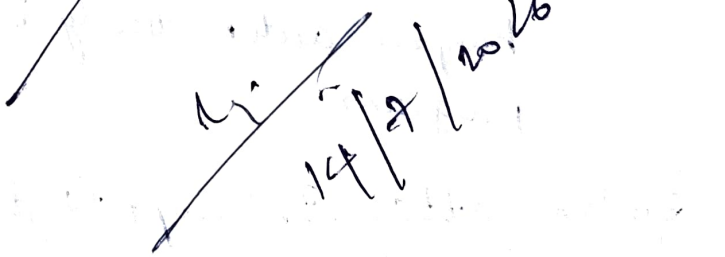
pulseIn () :- Reads a pulse on a pin.

syntax - pulseIn (pin, value)

PulseIn (pin, value, timeout).

Pulse In Long () :- It is an alternative to pulseIn

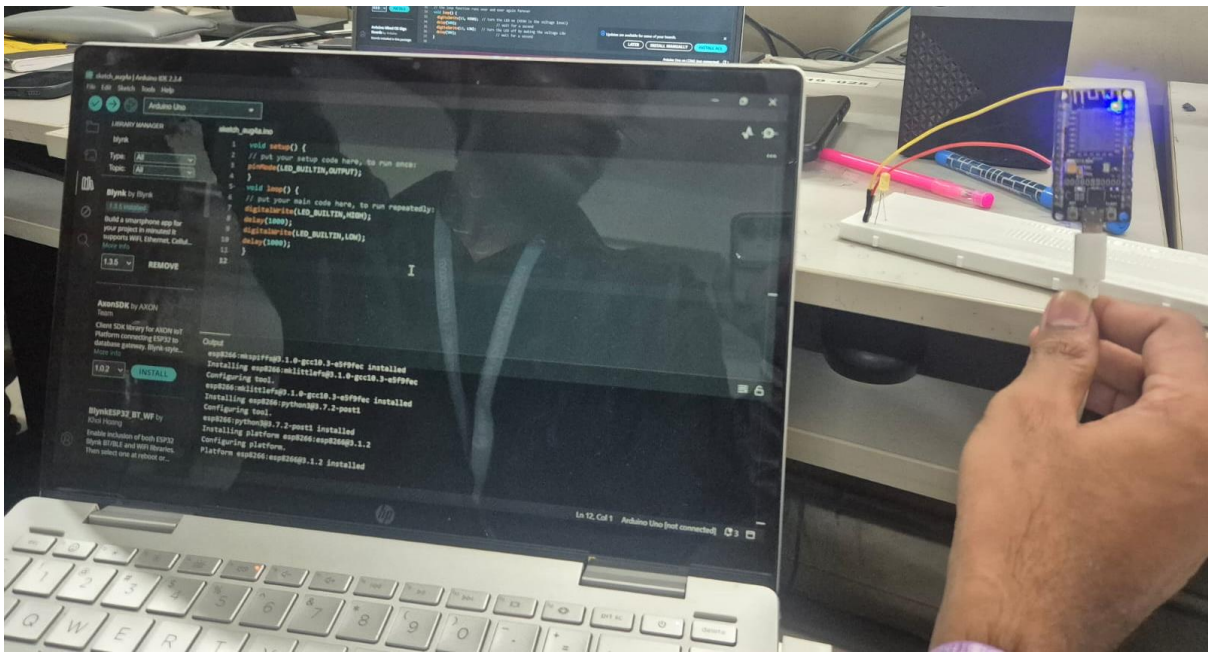
syntax — pulseIn long (pin, value).



BATCH 8

1. LED off board:

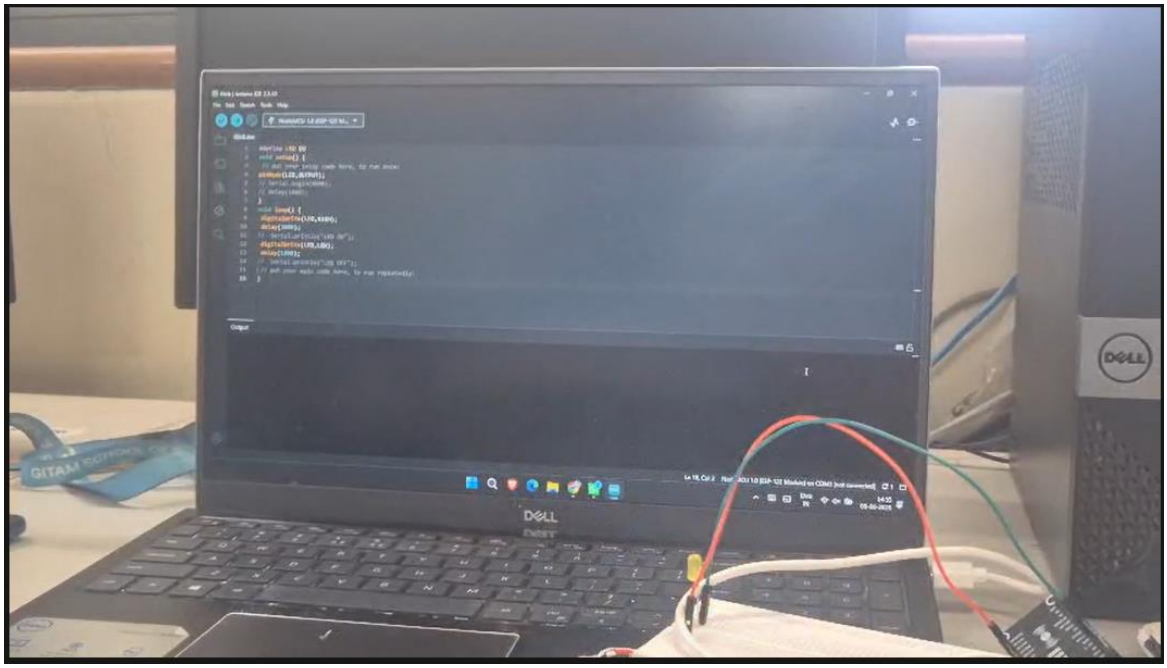
```
void setup()
{ // put your setup code here, to run once:
pinMode(LED_BUILTIN,OUTPUT);
}
void loop()
{ // put your main code here, to run repeatedly:
digitalWrite(LED_BUILTIN,HIGH);
delay(1000);
digitalWrite(LED_BUILTIN,LOW);
delay(1000);
}
```



Scan me!

2. LED on board:

```
void setup() {
 // put your setup code here, to run once: pinMode(D3,OUTPUT);
Serial.begin(9600); delay(1000);
}
void loop() {
digitalWrite(D3,HIGH);
delay(1000);
Serial.println("LED ON");
digitalWrite(D3,LOW);
delay(1000);
Serial.println("LED OFF"); // put your main code here, to run repeatedly:
 }
```



Scan me!